

3	Water is ejected from the nozzle of a hose at a speed of 2.0 m s^{-1} . The density of water is 1000 kg m^{-3} and the diameter of the nozzle is 0.50 cm . What is the force exerted on the nozzle by the ejected water?							
	A	0.039 N	B	0.079 N	C	0.31 N	D	7.9 N

Answer: B

Force **on water** = mass per unit time $\times$ change in velocity
 = (volume per unit time $\times$ density) $\times$ change in velocity
 = $(V/t)(\rho)\Delta v$
 = $(\pi r^2)(v)(\rho)(v)$
 = $\pi(0.0025^2)(2.0)(1000)(2.0) = 0.079 \text{ N}$

Magnitude of force **on nozzle** by water = Magnitude of force **on water** by nozzle = 0.079 N
 (Newton's 3rd law of motion)

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| 4 | A box resting on a table has two vertical and two horizontal forces acting on it as shown in the |
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